

Christopher Funk

Leah Eilers

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A Reflective Journey: Navigating My Cumulative Experience at Iowa State University

As an undergraduate student at Iowa State University, my experiences have equipped me with a toolbox of skills that I am confident will enable me to ethically and skillfully solve a wide range of electrical engineering problems.

Through the labs and projects at Iowa State, I have been exposed to and acquired a diverse range of skills that are directly applicable to my future profession. Through the various course activities at Iowa State, I have written code that was used for object distance and size detection using an IR sensor in an embedded robotics application; I have implemented a second-order discrete-time filter for IR sensor data in C in my signals and systems courses; and I have created and performed gate-level simulation on a working integrated circuit for a digital controller using Verilog and Cadence Virtuoso (the industry standard for mixed-signal IC design software).

The engineering training I have received as a student at Iowa State extends beyond theory and application of physics and math. Throughout courses like senior design and ethics, we were taught how to think critically about how the decisions we make as engineers impact society and the environment. Maintaining this big-picture perspective is particularly important for engineers working with today's increasingly interconnected and complex technologies.

In addition to training and hands-on experience, my studies at Iowa State have led me to engage with a variety of other students who are as passionate about engineering as I am. As an example, while taking EE230, I recall one discussion I had with another student while working through a homework on oscillators. The student asked me if I had a better grasp of why the Barkhausen Criteria are used as the necessary conditions for oscillation. What ensued was a passionate discussion regarding the origins of the

Barkhausen Criteria for oscillation and why they are necessary but not sufficient for oscillation. Due to this discussion, I still remember the Barkhausen Criteria and their application to oscillator circuits. This is just one example of the many discussions I have had with other students on the various concepts we have been exposed to through the lectures and coursework at Iowa State.

Going beyond lectures, the professors at Iowa State often point students to explore concepts beyond what is taught in the classroom; for example, in Signals and Systems II, our professor Shanda Moothadath frequently encouraged students to attend the research presentations of visiting lecturers who spoke on signal processing and machine learning concepts. Due to her prompting, I decided to attend a lecture regarding strategies for improving the accuracy of machine learning predictions. Iowa State also offers several valuable external resources with which students can expand their knowledge and experience beyond the classroom. One resource that I have used to expand my knowledge on various topics is Iowa State's subscription to IEEE Xplore. For example, part of my undergraduate research with Dr. Geiger involved working with a voltage reference. Since I had not taken any courses that cover voltage references, I set out to learn more about how these circuits work. Through Iowa State's IEEE Xplore subscription, I was able to gain access to an article by Dr. Behzad Razavi explaining the history and operation of bandgap voltage references. Having this type of high-quality information readily available on the internet is invaluable for students like me who want to expand on their knowledge beyond the classroom.

Given that there are so many eager engineering students at Iowa State, it is unsurprising that I have had the opportunity to engage with several passion-driven engineering student organizations. During my junior year, I had the opportunity to join Tau Beta Pi. In joining Tau Beta Pi, I was able to meet with other passionate students from a variety of engineering disciplines, and participating in organization events like Adopt-a-Highway which allowed me to give back to the community and take a break from coursework. Through my senior design project, I have also been able to join ChipForge, a club that enables students to design and fabricate integrated circuits using the sky130 fabrication process in a team environment. In addition to student organizations, attending Iowa State's Engineering Career Fair has also

allowed me to grow my networking skills in a low-pressure setting and, in the process, learn what engineers in the companies hiring Iowa State alumni are working on across the U.S.

In joining ChipForge, I have the opportunity to apply the knowledge I have gained as a student to develop novel solutions; in this case, my team and I have decided to create a microcontroller that supports various protocols needed for a micro quadcopter. Namely, it will feature support for nested PID acceleration and the various necessary communication protocols to interface with sensors common to a quadcopter application. As far as we are aware, no other open-source alternatives for this type of chip exist, so in creating this chip, we are taking the concepts learned in the various engineering courses and applying them to develop a novel microcontroller architecture.

Even though my experience as a student at Iowa State has been positive, there are a few things I would do differently if I had to redo my undergraduate studies. Firstly, I wish I had realized how critical it is for students to get as many of the general education classes out of the way before coming to study at Iowa State. With how competitive obtaining an internship has become, employers are looking for students who have completed higher-level engineering courses in their relevant disciplines and completed projects relevant to their studies. From my experience, these two factors are more relevant to employers than a student's grades. An additional argument for entering higher-level coursework sooner is that obtaining the skills from these higher-level engineering classes equips students with the background they need to work on more advanced projects. I believe that if I had been aware of exactly how this worked before beginning my journey at Iowa State, I would have made sure to set more time aside for projects that are relevant to electrical engineering.

Even though I do regret not having as many projects to challenge myself and showcase the full extent of the skills I have acquired, focusing on applying myself to the subjects in the classes I am in has enabled me to explore what areas in electrical engineering I enjoy. One subject I have enjoyed learning about is semiconductor physics, specifically, learning how to interpret semiconductor energy band diagrams and use them to intuit how the various semiconductor devices work when various electrical potentials are applied.

The breadth of my exposure has enabled my confidence in employing my engineering problem-solving skills to tackle a variety of problems. For example, during my summer 2026 internship at a large Midwestern electric utility company, I applied the programming skills I have acquired as a student to create a utility for automating the checking and updating of the data in a report containing relay settings. Before my internship, updating this report with the relay settings stored in the company's database was the responsibility of one of the engineers in the system protection and control department. Since it had settings for approximately three thousand relays, it was not feasible to thoroughly audit this report manually. One of the challenges in this particular project was ensuring that the correct settings fields were placed in the correct columns in the report, since the field names are not the same across the various relay models. To solve this problem in an adaptable and future-proof way, I chose to develop a custom file format based on .json, and then created an editor for this file format so that the engineers in the system protection department could easily set up how each relay type in the report should be mapped to the relay data in the database. In solving this problem, the naive approach would be to merely hardcode the mappings for the various relay types in the utility; however, this would have resulted in more development time, since I would have needed to debug every one of the many relay mappings to ensure that the report is receiving the correct data. With the data mapping generation tool I integrated into the utility, I abstracted the tedious task of manually defining the mappings to a user interface; this choice allowed for the rapid creation and debugging of new data mappings by the engineers once my internship was over.

Just like abstracting the problem of mapping data from one location to another enabled this relay settings report to be generated and checked faster, one of the most widely applicable skills that I have developed when it comes to solving complex engineering problems is the skill of creating and using abstraction. I have found that the first step in solving any engineering problem involves finding some way finding a way to break up a complex problem into manageable parts, these smaller problems can be further subdivided, until only simple problems are left. This iterative process is what I refer to as abstraction. To apply this skill, I try to always ask myself if solving the problem for a general case rather

than for the specific case I am dealing with would be more efficient. As a simple example, in the process of creating the optometrist perimeter controller using Verilog and Cadence Virtuoso, I first broke the design down into several high-level components: the decoder for flashing the light, the pseudo-random number generator, the counter for keeping track of the score, etc., and then I broke each of these subcomponents down into their required parts: the multiplexers, adders, decoders, and state machines needed to achieve the desired functionality.

In my engineering studies, I have gravitated towards anything involving circuit design, not because I feel that I have a special talent in this area, but because I find circuit design presents interesting challenges not encountered in other engineering disciplines; specifically, I feel that integrated circuit design requires a level of familiarity with applied physics concepts beyond that of other electrical engineering disciplines. Additionally, creating a functional integrated circuit is a rewarding experience, since bringing a circuit to the point where it is fabricable is an involved task.

Because of my passion for this electrical engineering sub-discipline, and because of this field's ever-developing nature, after graduating with my bachelor's, I plan on continuing my pursuit of knowledge in this sub-discipline by obtaining a master's specializing in Analog IC design. I would also like to continue my pursuit of building my knowledge in the Analog IC design field by entering a doctoral program if I have the opportunity; however, at this point in my journey as a student, I will see what opportunities are made available to me.

Regardless of whether I choose to pursue a doctoral degree after graduation, I know that through my various experiences as a student at Iowa State University, I have been shown the necessity of considering how my choices effect society and the world, and the importance of continually acquiring, improving, and adapting my knowledge as technology continues to advance.